Galaxy Image Classification - Quick Start README

Notebook: FDL_teamProci.ipynb | **Task:** 10-class Galaxy10 DECaLS morphology classification | **Framework:** TensorFlow/Keras

What this notebook does. It automatically downloads Galaxy10, uses a random 50% subset, resizes images to 128x128, creates stratified train/validation/test splits, normalizes and standardizes the images, augments the minority class, trains five CNN-based models, evaluates them, and produces Grad-CAM visual explanations.

1. Recommended environment

- **Python 3.11** (the notebook metadata records Python 3.11.13).
- **Kaggle or Google Colab with an NVIDIA GPU**; the original notebook was configured for a Tesla T4 with Internet enabled.
- A high-memory runtime is recommended. Training all five networks on CPU may be extremely slow.

2. Fastest setup: Kaggle / Colab

- Upload **FDL_teamProci.ipynb**.
- Enable a GPU accelerator and Internet access.
- Choose **Run all** and execute the cells from top to bottom.

The first cell installs **astroNN**. Galaxy10 data and ImageNet weights are downloaded automatically on first use.

3. Local installation

Run these commands in the folder containing the notebook:

```
python -m venv .venv
# macOS/Linux: source .venv/bin/activate #
Windows: .venv\Scripts\activate
python -m pip install --upgrade pip
pip install jupyter tensorflow astroNN numpy
pandas matplotlib seaborn scikit-learn opencv-
python pillow jupyter notebook
FDL_teamProci.ipynb
```

- Open the notebook and select **Kernel - Restart & Run All**.

4. Execution order and pipeline

- **Cells 0-10:** install/import packages, download data, one-hot encode labels, take the 50% sample, resize, split 70/15/15, normalize and standardize.
- **Cells 10-12:** augment only the least represented training class and create batches of 32 images.
- **Cells 15-41:** train and evaluate VGG16, DenseNet201, the custom CNN, ResNet50 and MobileNetV2. Maximum: 15 epochs; early stopping patience: 5.
- **Cells 43-45:** generate Grad-CAM explanations for VGG16, DenseNet201 and MobileNetV2. These cells require the corresponding models to have already been trained.

5. Files produced

- Best checkpoints in the current working directory: **best_VGG16.keras**, **best_DENSENET201.keras**, **best_custom1.keras**, **best_RESNET50.keras**, and **best_MobileNetV2.keras**.
- Inline notebook outputs: class-distribution plots, learning curves, classification reports, confusion matrices for VGG16/DenseNet201, and Grad-CAM overlays.

6. Important notes / troubleshooting

Out of memory: reduce `sample_frac` from 0.50 and/or `batch_size` from 32, then restart and rerun.

- **Results change between runs:** the random sampling step has no seed. For reproducibility, add `np.random.seed(42)`, `random.seed(42)`, and `tf.random.set_seed(42)` before sampling.
- **Download errors:** confirm Internet access; both Galaxy10 and pretrained ImageNet weights are fetched remotely.
- **Grad-CAM errors:** do not run the final cells in isolation after a kernel restart; retrain or load the relevant saved model first.

Optional headless execution

For a terminal-only run (training may take a long time):

```
jupyter nbconvert --to notebook --execute
FDL_teamProci.ipynb --output
FDL_teamProci_executed.ipynb
--ExecutePreprocessor.timeout=-1
```